

A SIMULTANEOUS EXPONENTIAL BASIS FOR TWO SUBSETS OF A FINITE ABELIAN GROUP

ABSTRACT. We answer Question 1.8 of Ferguson, Mayeli, and Sothanaphan affirmatively: every two nonempty subsets of equal cardinality in a finite abelian group admit a simultaneous exponential Riesz basis. More generally, two equal-norm tight frames with a common index set have a common set of indices, of cardinality the smaller frame dimension, selecting vectors that are linearly independent in both frames. The proof is a direct application of Edmonds' matroid intersection theorem.

1. THE RESULT

Let G be a finite abelian group with character group $\widehat{G}$. For $E \subseteq G$ and $\Lambda \subseteq \widehat{G}$, write

$$\mathcal{F}(E, \Lambda) = (\chi(x))_{x \in E, \chi \in \Lambda}.$$

Ferguson, Mayeli, and Sothanaphan asked whether every pair of nonempty equal-cardinality subsets of G has a simultaneous exponential Riesz basis [3, Question 1.8]:

Is it true that for every finite abelian group G , and every pair of nonempty subsets E_1, E_2 of G such that $|E_1| = |E_2|$, there exists a collection of exponentials $\Lambda \subseteq \widehat{G}$ such that the restrictions of the exponentials to E_1 form a Riesz basis for E_1 and the restrictions to E_2 form a Riesz basis for E_2 ?

The question was open as of the latest version of [3] (arXiv v6, 19 January 2021), which records that “the above question for two subsets is unresolved”; to our knowledge it has not been resolved since.

Theorem 1. *Let $E_1, E_2 \subseteq G$ be nonempty and satisfy $|E_1| = |E_2| = k$. Then there is a set $\Lambda \subseteq \widehat{G}$, with $|\Lambda| = k$, such that both $\mathcal{F}(E_1, \Lambda)$ and $\mathcal{F}(E_2, \Lambda)$ are nonsingular. Equivalently, the characters in Λ restrict to a Riesz basis of $L^2(E_i)$ for $i = 1, 2$.*

For cyclic G the existence of a simultaneous basis for an arbitrary family of subsets of equal size is due to Kolountzakis; see [3, Prop. 7.3], where the observation is credited to him and proved by taking $\Lambda = \{0, 1, \dots, k-1\}$ under the canonical identification of $\widehat{G}$ with G and evaluating a Vandermonde determinant. Theorem 1 removes the cyclicity hypothesis, at the cost of restricting to two subsets.

2020 *Mathematics Subject Classification.* Primary 43A40, 42C15; Secondary 05B35.

Key words and phrases. Finite abelian group, exponential Riesz basis, tight frame, matroid intersection.

2. A COMMON SET OF FRAME INDICES

Write $[n] = \{1, \dots, n\}$. For a matrix $X \in \mathbb{C}^{k \times n}$ and $A \subseteq [n]$, let X_A denote the submatrix formed by the columns indexed by A .

Proposition 2. *For $i = 1, 2$, let*

$$X_i = [x_{i1} \ \cdots \ x_{in}] \in \mathbb{C}^{k_i \times n}$$

satisfy

$$X_i X_i^* = \alpha_i I_{k_i}, \quad \|x_{ij}\|_2^2 = \frac{\alpha_i k_i}{n} \quad (j \in [n]),$$

where $\alpha_i > 0$. Then there is a set $B \subseteq [n]$, with $|B| = \min\{k_1, k_2\}$, such that the columns of both $X_{1,B}$ and $X_{2,B}$ are linearly independent.

Proof. Let M_i be the linear matroid on $[n]$ represented by the columns of X_i , and write

$$r_i(A) = \text{rank } X_{i,A}.$$

Since $X_i X_i^* = \alpha_i I_{k_i}$, the rank of M_i is k_i . For $A \subseteq [n]$, set

$$Q_i(A) = X_{i,A} X_{i,A}^*.$$

Since

$$X_i X_i^* - Q_i(A) = X_{i,[n] \setminus A} X_{i,[n] \setminus A}^* \succeq 0,$$

we have

$$0 \preceq Q_i(A) \preceq X_i X_i^* = \alpha_i I_{k_i}.$$

The matrix $Q_i(A)$ has rank $r_i(A)$, and each of its nonzero eigenvalues is at most α_i . Hence

$$\alpha_i r_i(A) \geq \text{tr } Q_i(A) = \sum_{j \in A} \|x_{ij}\|_2^2 = \frac{\alpha_i k_i |A|}{n},$$

so

$$(1) \quad r_i(A) \geq \frac{k_i |A|}{n}.$$

Assume without loss of generality that $k_1 \leq k_2$. Edmonds' matroid intersection theorem [2] gives

$$\max\{|I| : I \text{ is independent in both } M_1 \text{ and } M_2\} = \min_{A \subseteq [n]} (r_1(A) + r_2([n] \setminus A)).$$

Using (1), for every $A \subseteq [n]$ we have

$$r_1(A) + r_2([n] \setminus A) \geq \frac{k_1 |A| + k_2 (n - |A|)}{n} \geq k_1.$$

Thus there is a common independent set of size k_1 , which is the largest possible size in M_1 . $\square$

Remark 3. The inequalities (1) are the uniform-density conditions for the two column matroids: a loopless matroid on n elements, of rank r and with rank function $A \mapsto r(A)$, is *uniformly dense* if $r(A) \geq r|A|/n$ for every subset A of its ground set, equivalently if the density $\rho(A) = |A|/r(A)$ never exceeds the density n/r of the ground set itself. This is condition (2) of [1, Theorem 1.1]. The proof above applies verbatim to show that two uniformly dense matroids on a common ground set have a common independent set whose size is the smaller of their ranks.

The passage from an equal-norm tight frame to a uniformly dense column matroid is the complex counterpart of one of the applications of that theorem in [1], where a real representable matroid is shown to admit a projection representation with constant diagonal equal to the reciprocal of its density precisely when it is strictly uniformly dense in the sense used there. Proposition 2 is thus not offered as new in isolation; the new content is the resolution of [3, Question 1.8].

3. PROOF OF THE SIMULTANEOUS-BASIS THEOREM

Proof of Theorem 1. Set $n = |G| = |\widehat{G}|$. For $E \subseteq G$, $|E| = k$, let

$$F_E = (\chi(x))_{x \in E, \chi \in \widehat{G}} \in \mathbb{C}^{k \times n}.$$

Character orthogonality gives

$$(F_E F_E^*)_{x,y} = \sum_{\chi \in \widehat{G}} \chi(x) \overline{\chi(y)} = n \delta_{x,y},$$

and therefore $F_E F_E^* = n I_k$. Every column of F_E has squared norm k , since $|\chi(x)| = 1$. Thus F_E satisfies Proposition 2 with $\alpha = n$.

Apply the proposition to F_{E_1} and F_{E_2} , with their columns indexed by $\widehat{G}$. It yields a set $\Lambda \subseteq \widehat{G}$ of cardinality k for which both $\mathcal{F}(E_1, \Lambda)$ and $\mathcal{F}(E_2, \Lambda)$ have linearly independent columns. These square matrices are therefore nonsingular. In finite dimension every basis is a Riesz basis. $\square$

Remark 4. The theorem is sharp with respect to the number of subsets: three two-element subsets of $\mathbb{Z}_2^2$ need not admit a simultaneous basis [3, Example 7.2], where this case is likewise credited to Kolountzakis. The argument above is qualitative and gives no uniform bound on the condition numbers of the two Fourier submatrices. It also applies verbatim to any two row restrictions of equal size in a complex Hadamard matrix normalized by $HH^* = n I_n$.

REFERENCES

- [1] K. Devriendt and R. Mulas, *Uniform density in matroids, matrices and graphs*, Australas. J. Combin. **94** (2026), no. 1, 145–176.
- [2] J. Edmonds, *Matroid intersection*, Ann. Discrete Math. **4** (1979), 39–49, doi:10.1016/S0167-5060(08)70817-3.
- [3] S. Ferguson, A. Mayeli, and N. Sothanaphan, *Riesz bases of exponentials and multi-tiling in finite abelian groups*, arXiv:1904.04487v6 (2021).